

Data Mining and Knowledge Discovery

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14/09/2026

Acknowledgement
These slides are reused **as-is from previous years**, prepared by Dr. Huseyin Kusetogullari, DIDA, BTH.

Outline

- Data Mining (why?), (what?)
- Data Mining Process
  - ✓ The KDD Process (Fayyad, 1996)
  - ✓ CRISP-DM (Shearer, 2000)
  - ✓ OSEMN
- Steps for Successful Data Mining Process
- Regression and Classification
- Clustering and Outlier Detection
- Regression Models
  - Linear Model
  - Example

Data Mining - From data to insights

- Data Mining is the process of extracting knowledge and insights from (big) data.

- Important steps of the Data mining process are:
  - Acquire and clean the data.
  - Extract valuable insights using machinelearning.
  - Present/Visualize the gathered insights.

Why Data Mining?

Why Data Mining is important?

- Lots of data is being collected and warehoused
  - Web data, e-commerce
  - purchases at department/ grocery stores
  - Bank/Credit Card transactions







- Computers have become cheaper and more powerful

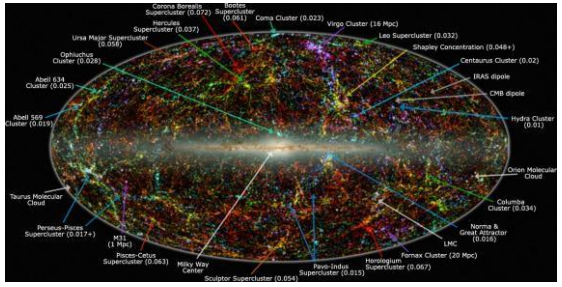
Some Examples:



- Direct marketing:
  - Goal: Increase the profit by targeting a set of customers likely to buy a new cell-phone product.



- Fraud Detection
  - Goal: Predict fraudulent cases in credit card transactions.



- Sky Survey Cataloging
  - Goal: To predict class (star or galaxy) of sky objects, especially visually faint ones, based on the telescopic survey images (from Palomar Observatory).
    - 3000 images with 23,040 x 23,040 pixels per image.

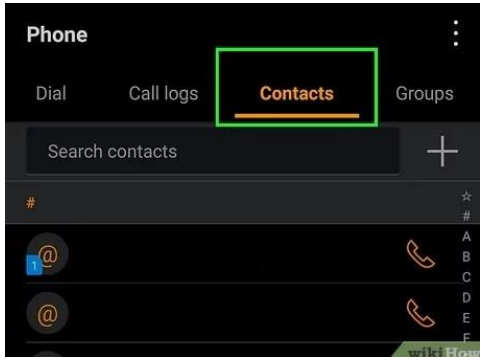
What is and not Data Mining?

What is not Data Mining?

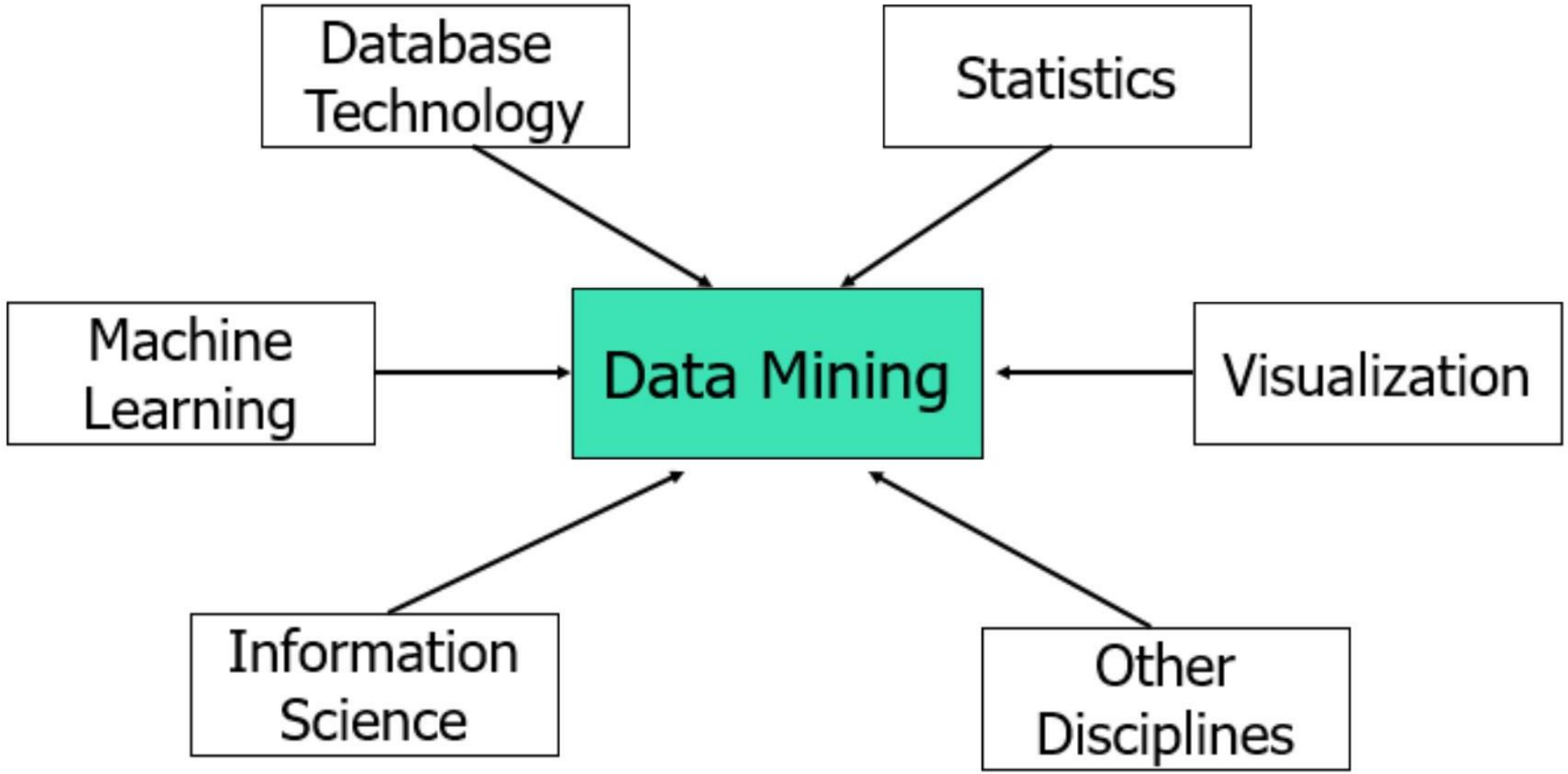
- Look up phone number in phone directory
- Query a Web search engine for information about "Amazon"

What is Data Mining?

- Certain names are more prevalent in certain US locations (O'Brien, O'Rurke, O'Reilly… in Boston area)
- Group together similar documents returned by search engine according to their context (e.g. Amazon rainforest, Amazon.com,)



Day	outlook	temperature	humidity	windy	play
1	sunny	85	85	false	no
2	sunny	80	90	true	no
3	overcast	83	86	false	yes
4	rainy	70	96	false	yes
5	rainy	68	80	false	yes
6	rainy	65	70	true	no
7	overcast	64	65	true	yes
8	sunny	72	95	false	no
9	sunny	69	70	false	yes
10	rainy	75	80	false	yes
11	sunny	75	70	true	yes
12	overcast	72	90	true	yes
13	overcast	81	75	false	yes
14	rainy	71	91	true	no


Database Technology
Statistics
Machine Learning
Data Mining
Visualization
Information Science
Other Disciplines

What Is Data Mining?



- Data mining (knowledge discovery from data)
  - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
  - Data mining: a misnomer?
- Alternative names
  - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- Watch out: Is everything “data mining”?
  - Simple search and query processing
  - (Deductive) expert systems



Data Mining process

- There are several process standards describing the datamining process:
  - The KDD Process (Fayyad, 1996)
  - CRISP-DM (Shearer, 2000)
  - OSEMN

The KDD process

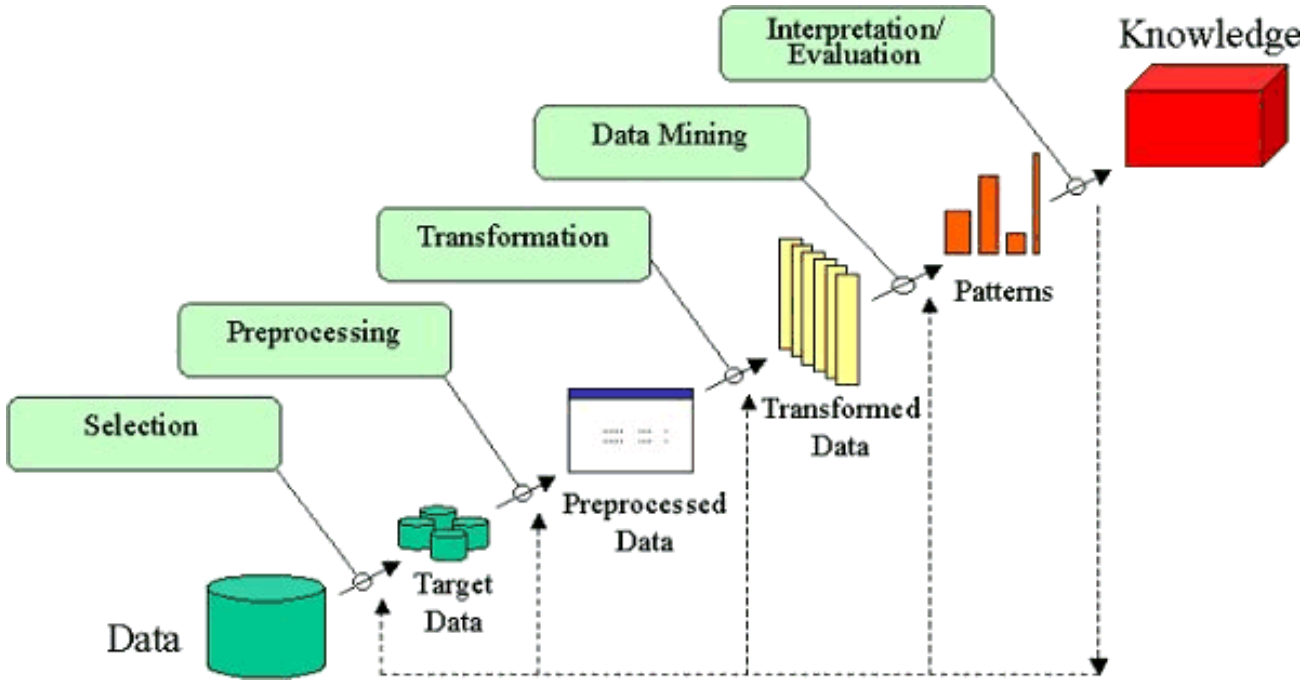
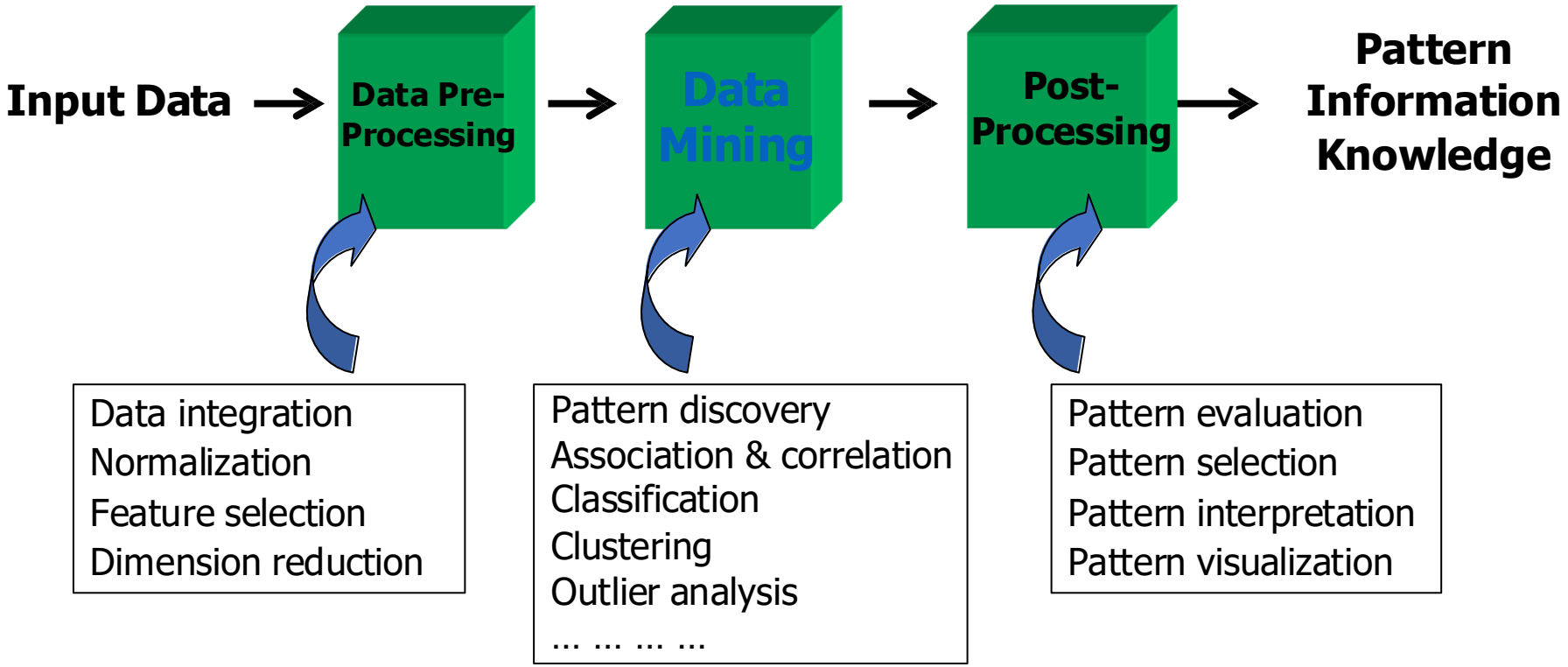


Figure: **K**nowledge **D**iscovery in **D**atabases.

KDD Process: A Typical View from ML and Statistics



- This is a view from typical machine learning and statistics communities

CRISP-DM

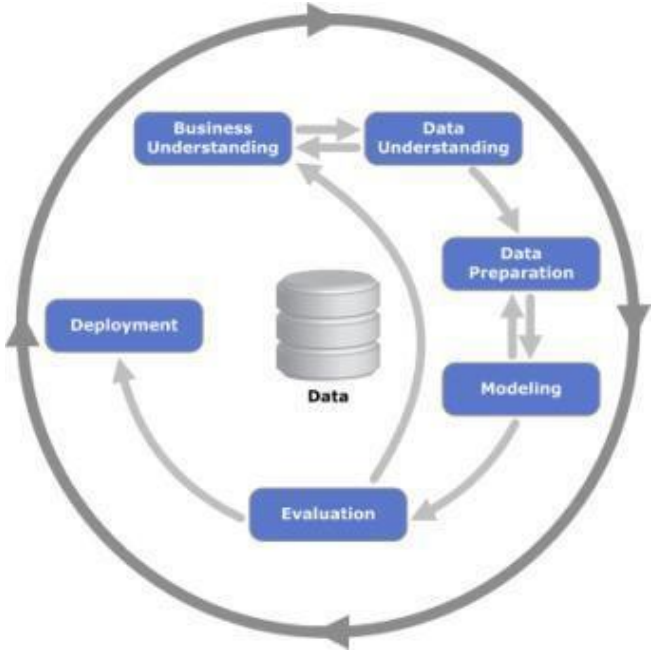


Figure: **CR**oss **I**ndustry **P**rocess for**D**ata **M**ining

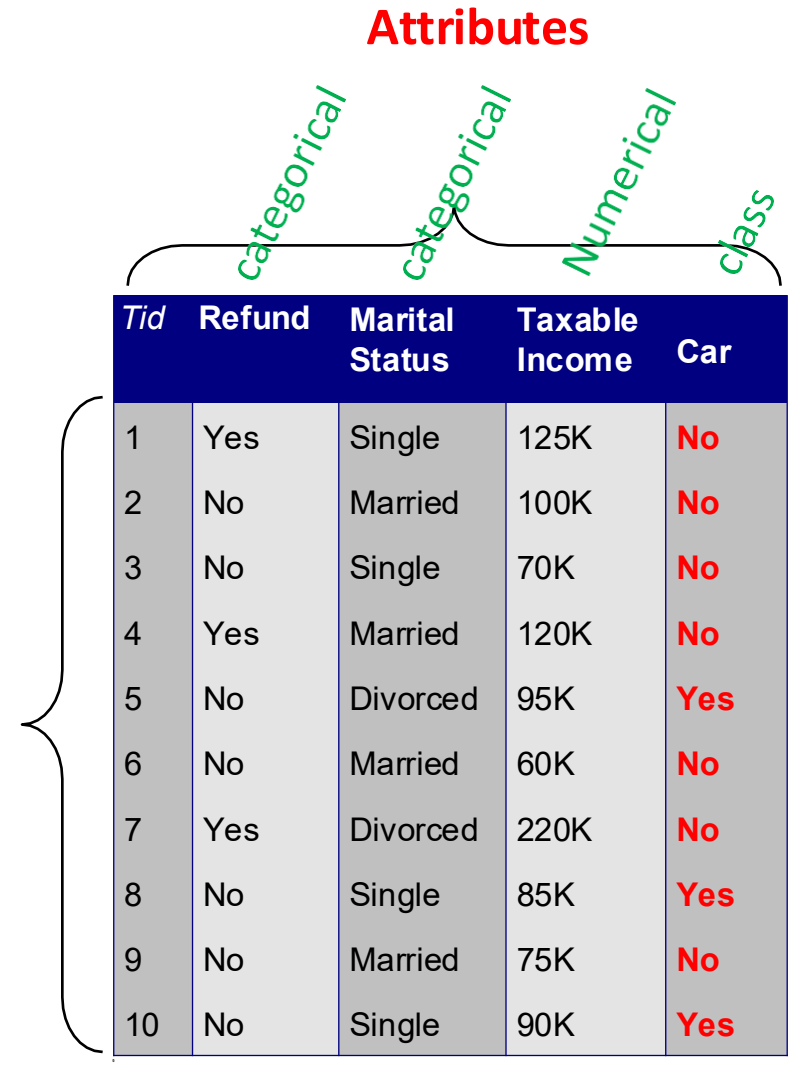
OSEMN



Figure: **O**btain data, **S**crubb data, **E**xplore data, **M**odel, i**N**terpret result.

What is Data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance



Tid	Refund	Marital Status	Taxable Income	Car
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute Values

- Attribute values are numbers or symbols assigned to an attribute
- Distinction between attributes and attribute values

  - Same attribute can be mapped to different attribute values
    - Example: height can be measured in feet or meters
  - Different attributes can be mapped to the same set of values
    - Example: Attribute values for ID and age are integers
    - Attribute values may have limitations
      - ID has no limit but age has a maximum and minimum value

Types of Attributes

- There are different types of attributes
  - Nominal
    - Examples: ID numbers, eye color, zip codes
  - Ordinal
    - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
  - Interval
    - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
  - Ratio
    - Examples: temperature in Kelvin, length, time, counts

Data Quality



What kinds of data quality problems?



How can we detect problems with the data?



What can we do about these problems?



Examples of data quality problems:

Noise and outliers , missing values, duplicate data

Data Integration



Data integration:

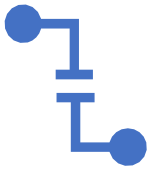
Combines data from multiple sources into a coherent store



Schema integration

Integrate metadata from different sources

Entity identification problem: identify real world entities from multiple data sources, e.g., A.cust-id ≡ B.cust-#

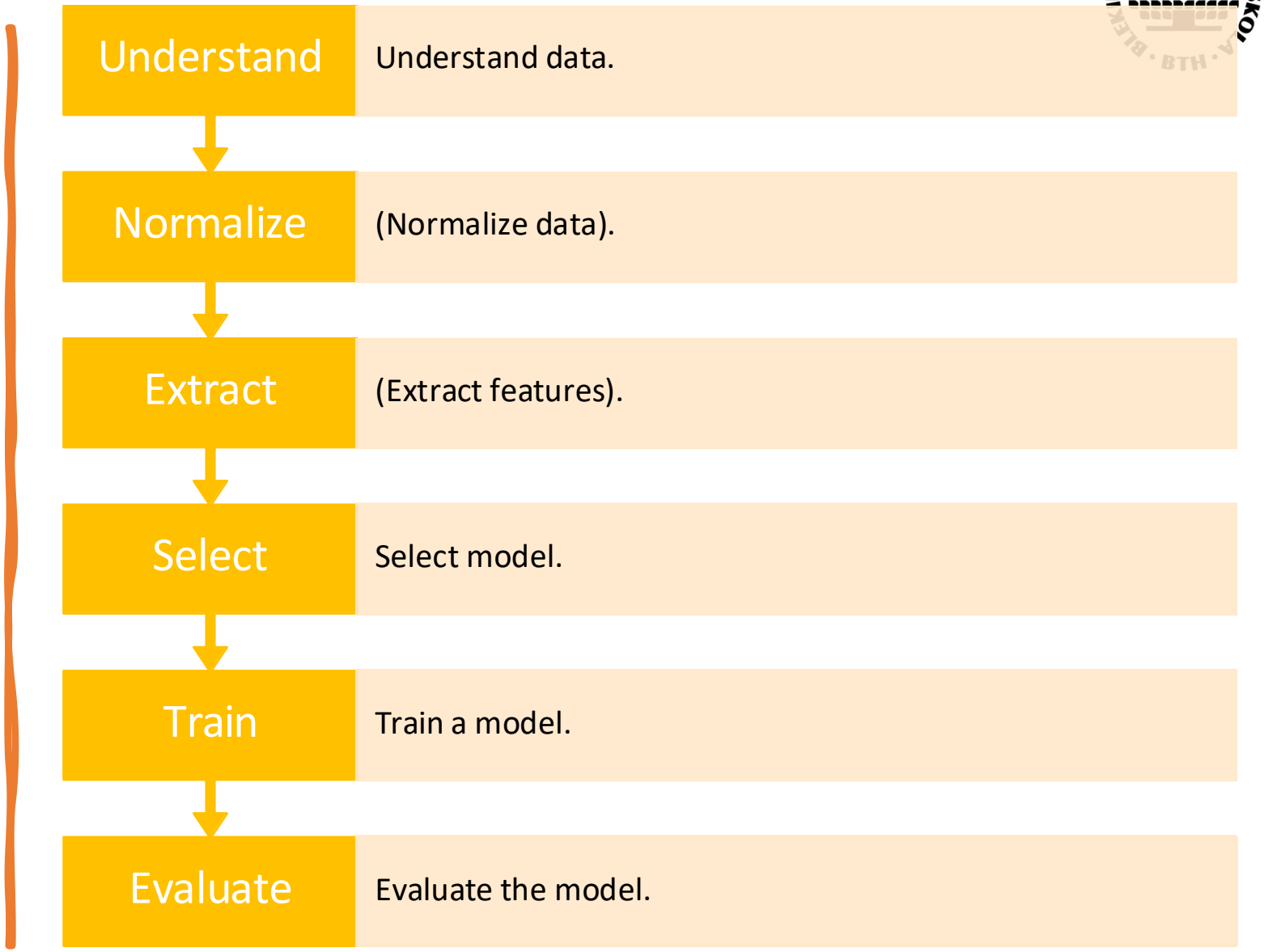


Detecting and resolving data value conflicts

for the same real-world entity, attribute values from different sources are different

possible reasons: different representations, different scales, e.g., metric vs. British units

Steps for performing successful data-mining



Starting with data mining



All data science applications start with data.



Therefore, if you want to do successful in data science you first need to understand the data.



Ask yourself: A given data problem solvable with any data science approach? if the answer is no, then the problem is probably insolvable with machine learning algorithms.

Understand the Relationship of Data

- Given the temperature over the last day, could we predict the temperature for the next hour?
- Given the number of ice-creams sold, could we predict the number of cars sold that day?
- Given the number of students admitted to BTH, could we predict the weather tomorrow?

Understand the Relationship of Data

- Given the temperature over the last day, could we predict the temperature for the next hour?
- Yes. There is a connection between the two.

- Given the number of ice-creams sold could we predict the number of cars sold that day?
- No. There are no connection.

- Given the number of students admitted to BTH could we predict the weather tomorrow?
- No. There are no connection.

The scale of the data

- Data can be stored in formats with different scales.

- The temperature could for example be stored as the mean
- Over a second, minute, day, week, month or year.

- Data can be aggregated to a form with a larger unit. For example, the mean daily temperature can be average into the mean weakly temperature.



The scale of the data vs the scale of a prediction

- How good could we predict the mean temperature tomorrow from the last years temperature data stored as the mean temperature per *X*. Where *X* is:
  - Minute,
  - Day,
  - Week,
  - Month,
  - Year,

The scale of the data vs the scale of a prediction

- How good could we predict the mean temperature tomorrow from the last years temperature data stored as the mean temperature per *X*. Where *X* is:
  - Minute, Okay
  - Day, Okay
  - Week, Okay
  - Month, Starting to get hard
  - Year, Really hard

Dimensionality reduction

- The curse of dimensionality:
  - The computational cost often grows quadratic with input size.
  - Data becomes increasingly sparse in the space it occupies.

- Lower the number of dimensions (but keep as much of the variation/information as possible)
  - Principal component analysis (PCA).
  - Singular value decomposition (SVD).

Major Tasks in Data Preprocessing



DATA CLEANING



MISSING VALUES



BINNING



DATA REDUCTION



DATA TRANSFORMATION

1. Data cleaning

- Data cleaning is the process to remove incorrect data, incomplete data and inaccurate data from the datasets, and it also replaces the missing values.

	No.		
Mobile Phones	1	Apple	No
	2	Samsung	Yes
	3	**BMW**	**Yes**
	4	Apple	No
	5	**Ford**	**Yes**



	No.		
Mobile Phones	1	Apple	No
	2	Samsung	Yes
	4	Apple	No

Data Cleaning

Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error

incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data e.g., Occupation=“ ” (missing data)

noisy: containing noise, errors, or outliers e.g., Salary=“−10” (an error)

inconsistent: containing discrepancies in codes or names, e.g.,

- Age=“42”, “Fourtytwo”
- Birthday=“03/07/2010”, “3rd July 2010”
- rating “1, 2, 3”, now rating “A, B, C”



Example for Data Cleaning or Standardization

2. Data Editing

- We can perform a Data editing and change the phones.
- We edit the dataset to have a clearer analysis of the data when performing data manipulation in the later steps within the data science process.

	No.		
Mobile Phones	1	Apple	No
	2	Samsung	Yes
	3	**BMW**	**Yes**
	4	Apple	No
	5	**Ford**	**Yes**



	No.		
Mobile Phones	1	Apple	No
	2	Samsung	Yes
	3	Samsung	**Yes**
	4	Apple	No
	5	Apple	**Yes**

3. Missing values

- It is very much usual to have missing values in your dataset.
- It may have happened during data collection, or maybe due to some data validation rule, but regardless missing values must be taken into consideration.
  - Deleting Rows with missing values.
  - Impute missing values for categorical variable.
  - Prediction of missing values





Weighting and averaging



Filling with the Nearest data

Pre-processing: Binning

- Sometimes it can help to bin variables into discrete classes.

- For example length into the classes: small, medium, large.

Pre-processing: Normalization

- Data normalization is a technique used in data mining to transform the values of a dataset into a common scale.
- This is important because many machine learning algorithms are **sensitive** to the scale of the input features
- Machine learning algorithms can produce better results with the normalized data.

Normalization Methods

There are several different normalization techniques that can be used in data mining, including:

1. **Min-Max normalization:** This technique scales the values of a feature to a range between 0 and 1. This is done by subtracting the minimum value of the feature from each value, and then dividing by the range of the feature.
2. **Z-score normalization:** This technique scales the values of a feature to have a mean of 0 and a standard deviation of 1. This is done by subtracting the mean of the feature from each value, and then dividing by the standard deviation.
3. **Decimal Scaling:** This technique scales the values of a feature by dividing the values of a feature by a power of 10.
4. **Logarithmic transformation:** This technique applies a logarithmic transformation to the values of a feature. This can be useful for data with a wide range of values, as it can help to reduce the impact of outliers.
5. **Root transformation**: This technique applies a square root transformation to the values of a feature. This can be useful for data with a wide range of values, as it can help to reduce the impact of outliers.

* It is important to note that normalization should be applied only to the input features, not the target variable, and that different normalization technique may work better for different types of data and models.

Examples

- Min-Max normalization:

$$v' = \frac{v - \min_A}{\max_A - \min_A}(new_\max_A - new_\min_A) + new_\min_A$$

Salary	Normalized Salary
100000	1
52000	0.5
28000	?
12000	?
4000	?

- Z-score normalization:

$$x_{scaled} = \frac{x - mean}{sd} \longrightarrow s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

mean = 39200
sd = 38616.059

Salary	Normalized Salary
100000	1.5744
52000	0,3314
28000	?
12000	?
4000	?

Advantages and Disadvantages

- **Advantages**:

1. **Improve performance of machine learning algorithms:** Normalization can help to improve the performance of machine learning algorithms by scaling the input features to a common scale. This can help to reduce the impact of outliers and improve the accuracy of the model.
2. **Better handling of outliers:** Normalization can help to reduce the impact of outliers by scaling the data to a common scale, which can make the outliers less influential.
3. **Improved interpretability of results:** Normalization can make it easier to interpret the results of a machine learning model.
4. **Better generalization:** Normalization can help to improve the generalization of a model, by reducing the impact of outliers and by making the model less sensitive to the scale of the inputs.

- **Disadvantages**:

1. **Loss of information**: Normalization can result in a loss of information if the original scale of the input features is important.
2. **Impact on outliers:** Normalization can make it harder to detect outliers as they will be scaled along with the rest of the data.
3. **Additional computational costs:** Normalization can add additional computational costs to the data mining process, as it requires additional processing time to scale the data.

Data Mining Functions: Classification and Prediction



Classification and prediction

Construct models (functions) based on some training examples

Describe and distinguish classes or concepts for future prediction

- E.g., classify countries based on (climate), or classify cars based on (size)

Predict some unknown or missing numerical values



Typical methods

Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, …



Typical applications:

Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, …

Classification



Classification is an important task in data mining. It involves assigning a class label to each instance in a dataset based on its features. (Training Data)

The goal of classification is to build a model that accurately predicts the class labels of new instances based on their features.

There are two main types of classification: binary classification and multi-class classification.

- Binary classification involves classifying instances into two classes, such as “spam” or “not spam”, while
- multi-class classification involves classifying instances into more than two classes

Data Mining Functions: Cluster and Outlier Analysis

Cluster analysis

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications

Outlier analysis

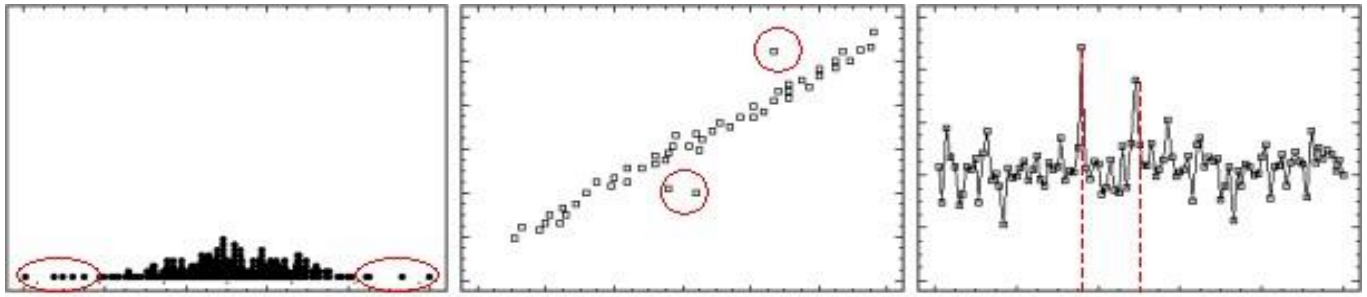
- Outlier: A data object that does not comply with the general behavior of the data
- Noise or exception? — One person's garbage could be another person's treasure
- Methods: by product of clustering or regression analysis, …

Outlier Detection

- To find exceptional data in various datasets and uncover the implicit patterns of rare cases
- Inherent variability - reflects the natural variation
- Measurement error (inaccuracy and mistakes)
- Long been studied in statistics
- An active area in data mining in the last decade
- Many applications
  - Detecting credit card fraud
  - Discovering criminal activities in E-commerce
  - Identifying network intrusion
  - Monitoring video surveillance and others

Outliers are Everywhere

- Data values that appear inconsistent with the rest of the data.
- Some types of outliers

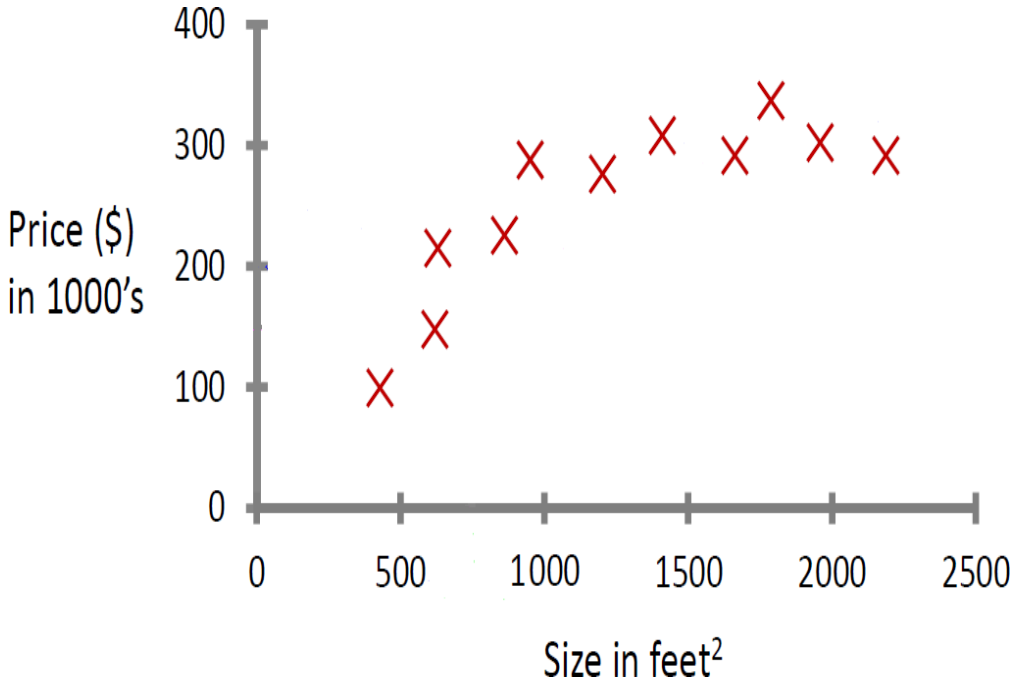


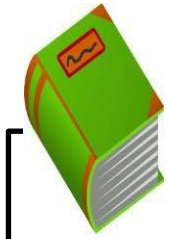
(a) Outliers w.r.t. a distribution (b) Outliers w.r.t. a pattern (c) time series outliers

Regression and Linear Models

What is Regression?

Housing price prediction (regression)





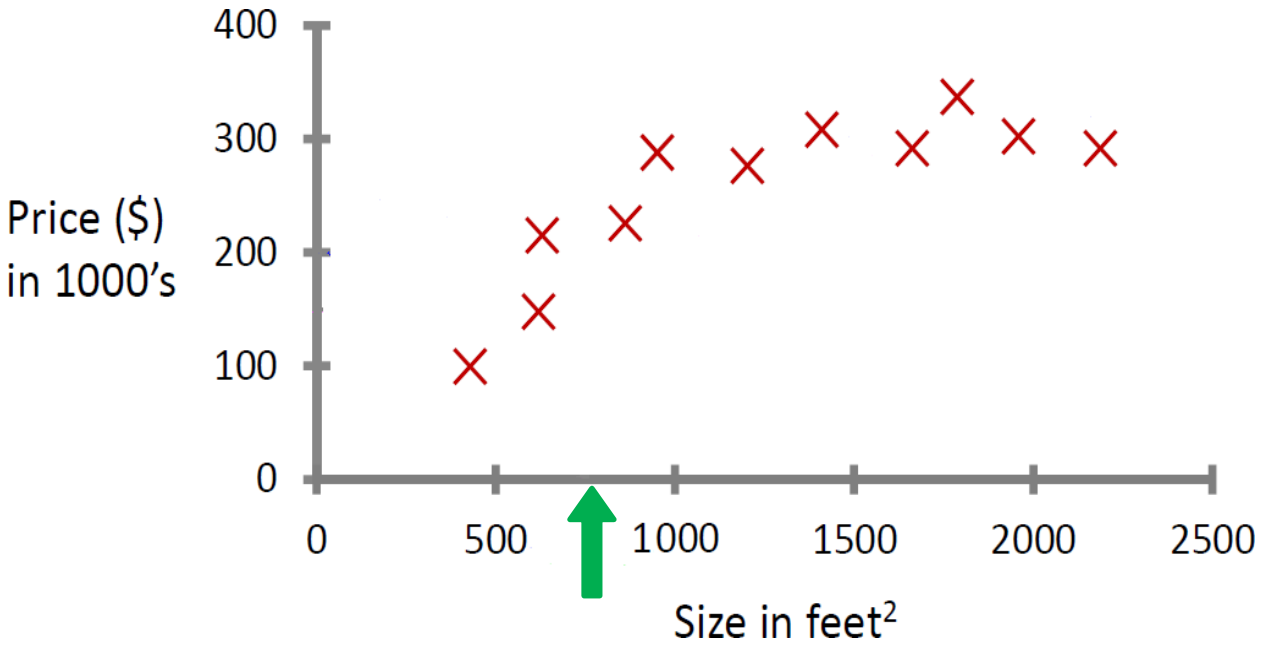
Definition: Regression

is a form of predictive modelling technique which investigates the relationship between a dependent variable[Output] and several independent variables[Inputs]

What is Regression?

Housing price prediction (regression)

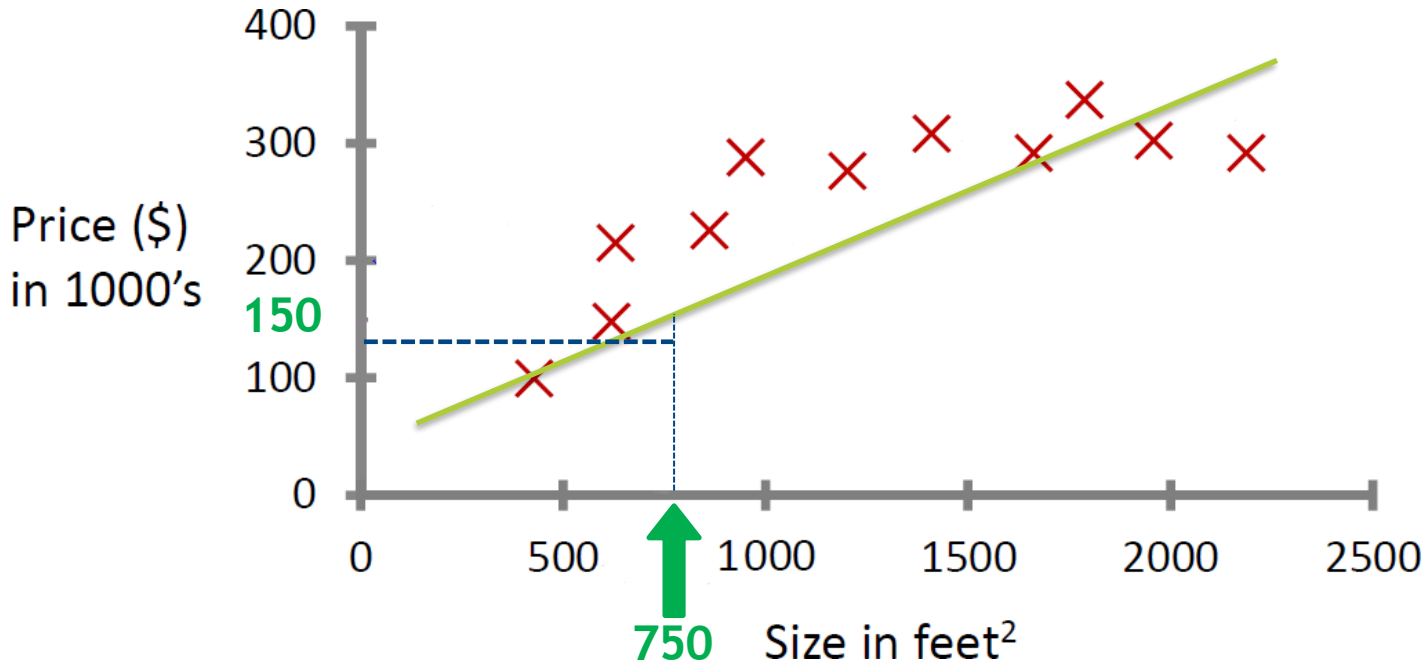
- Suppose that you want to sell a house of **size 750 feet²** and want to know how much you can get for this house, i.e. **predict its price**.
- How can a learning algorithm help you?



What is Regression?

Housing price prediction (regression)

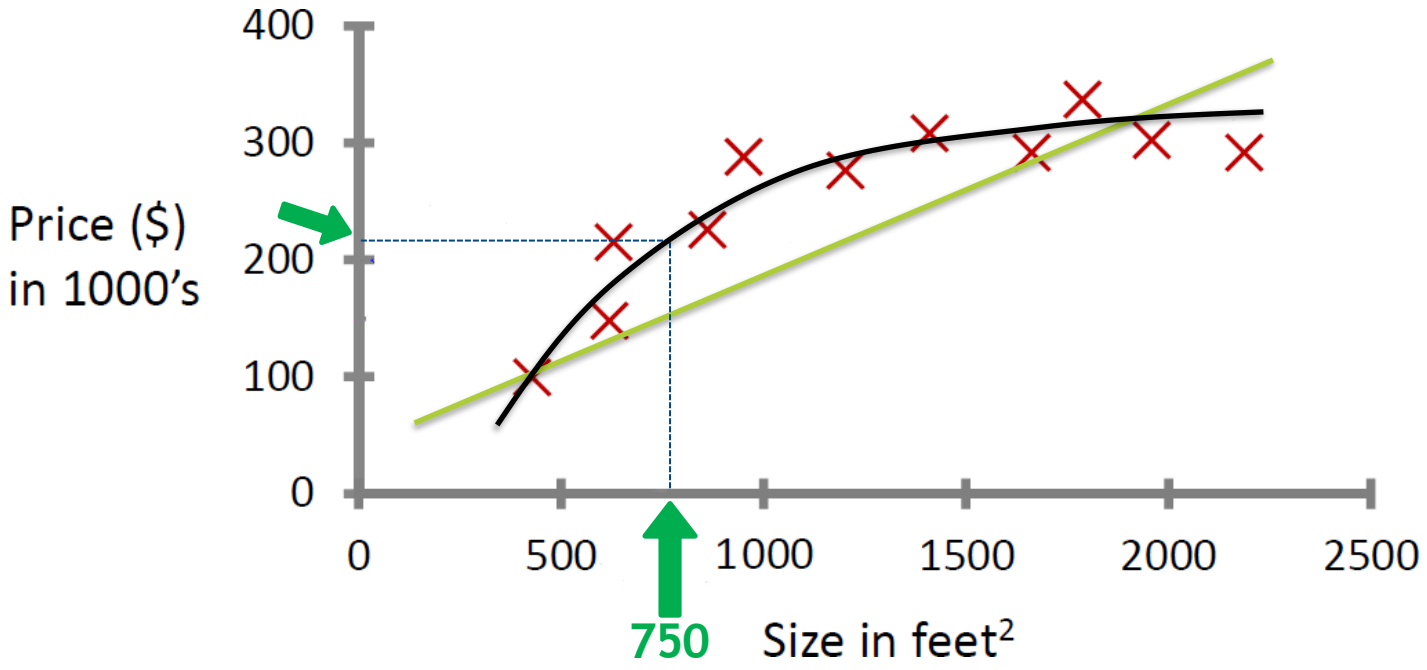
- you can fit astraight line to the data and predict the price of the house.



What is Regression?

Housing price prediction (regression)

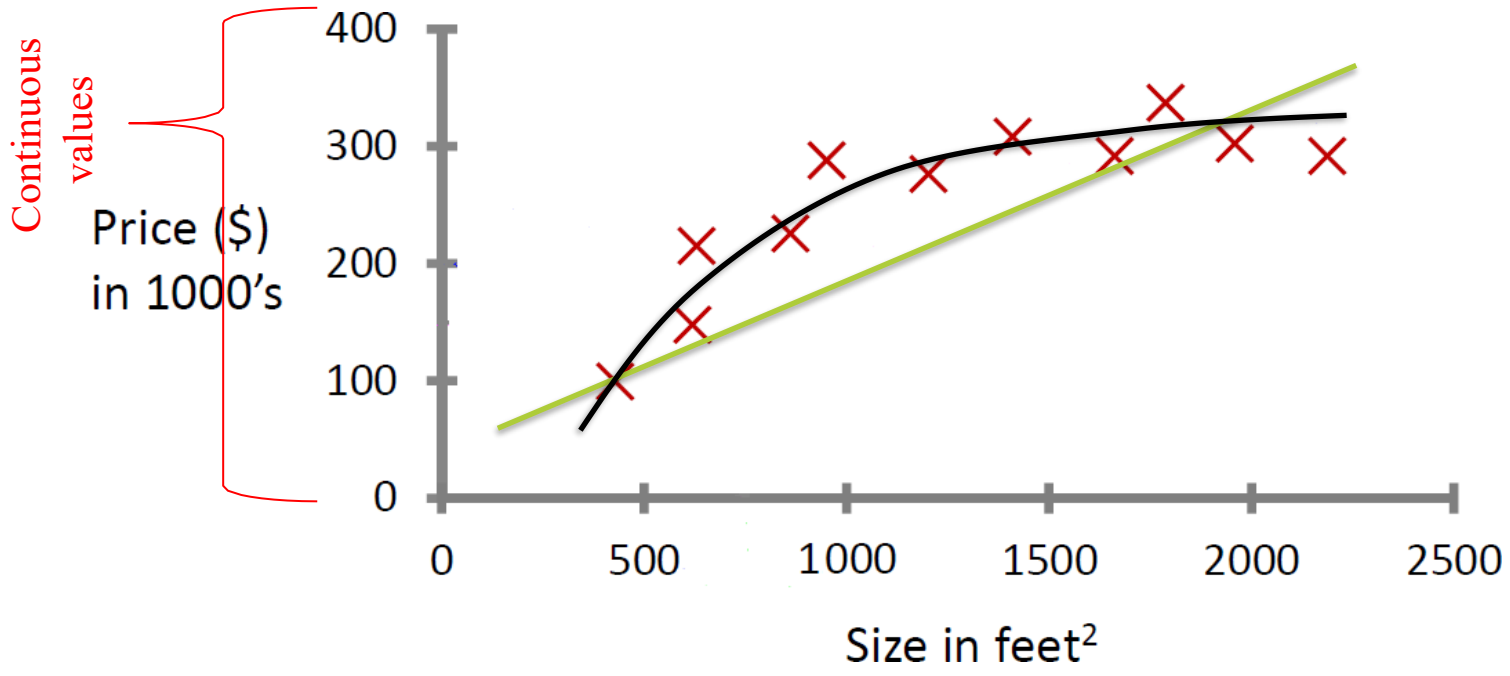
- you can fit astraight line to the data and predict the price of the house.
- or maybe it's better to fit aquadratic function (2nd-order polynomial).



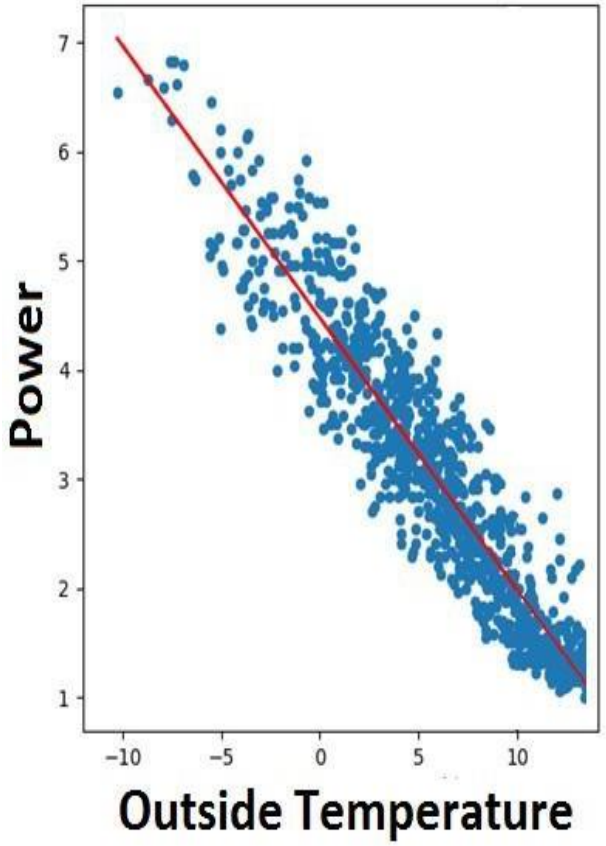
What is Regression?

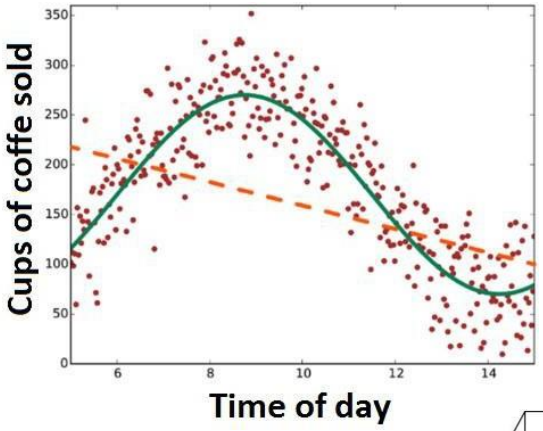
Housing price prediction (regression)

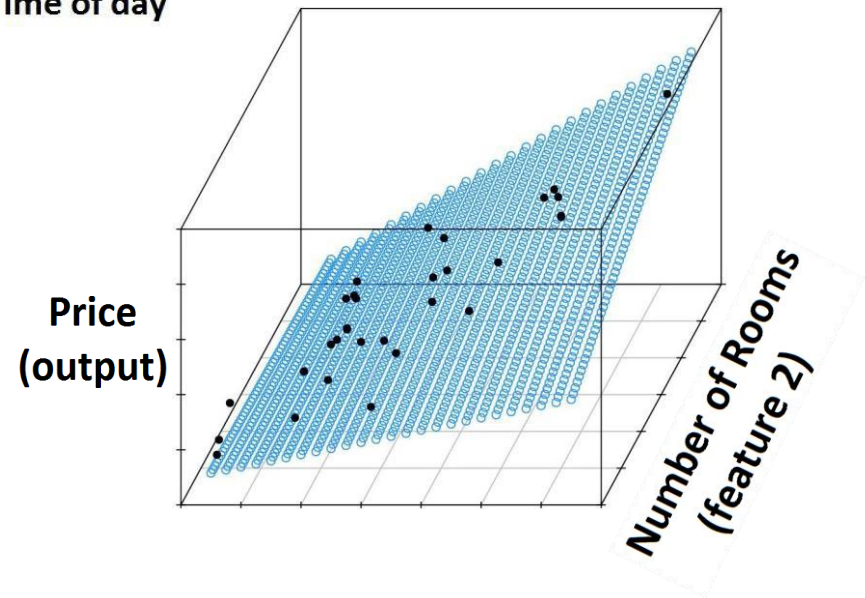
- This is an example of a **supervised learning** algorithm:
  - The right answers (here, the prices) are given in the training dataset.
- More specifically, this example was a **regression problem**:
  - Predicting a continuous valued output (price).



More regression examples







Linear Models

- Linear models are referred to the geometric models.
  - It could be described in 2-dimensional.
  - It could be described in 3-dimensional.
  - or in d-dimension.
- Supervised based methods.
- Linear models help to find the regression line which show the linear relationship between 2 or more than 2 variables.
- Linear model provides
  - Classification
  - Regression

Linear Models

- There is a distinction between **classification** and **regression**, two particular instances of supervised learning.

  - **Classification** is when our labels yy can only take a finite set of values (categories or classes). Examples include:
    - Handwritten digit recognition: xx is an image with a handwritten digit, yy is a digit between 0 and 9.
    - Spam filtering: xx is an e-mail, and yy is 0 or 1 whether that e-mail is a spam or not.

  - **Regression** is when our labels yy can take any real (continuous) value. Examples include:
    - Predicting stock market.
    - Predicting sales.
    - Detecting the age of a person from a picture.

Univariate Linear Regression

- The regression line summarizes the linear relationship between two different variables. Regression coefficient or slope b' is found as follows

$$b = \frac{\sigma_{xy}}{\sigma_{xx}} = \frac{cov(x,y)}{var(x)} = \frac{\frac{1}{n-1}\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1}\sum_i^n (x_i - \bar{x})^2}$$

$\sigma_{xx} = \sigma_x{}^2$

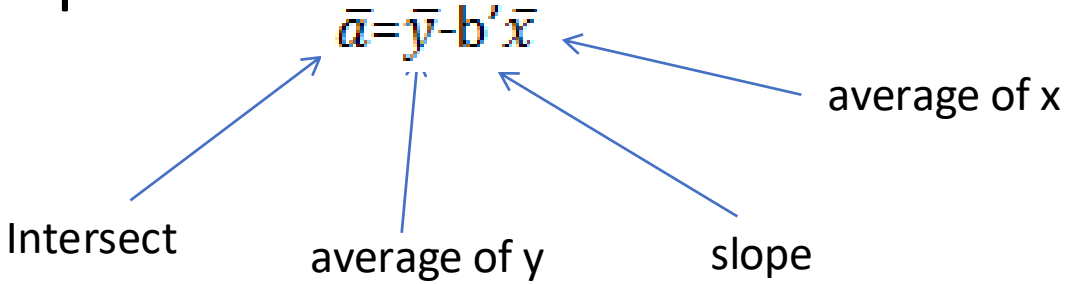
(Variance of variable x)

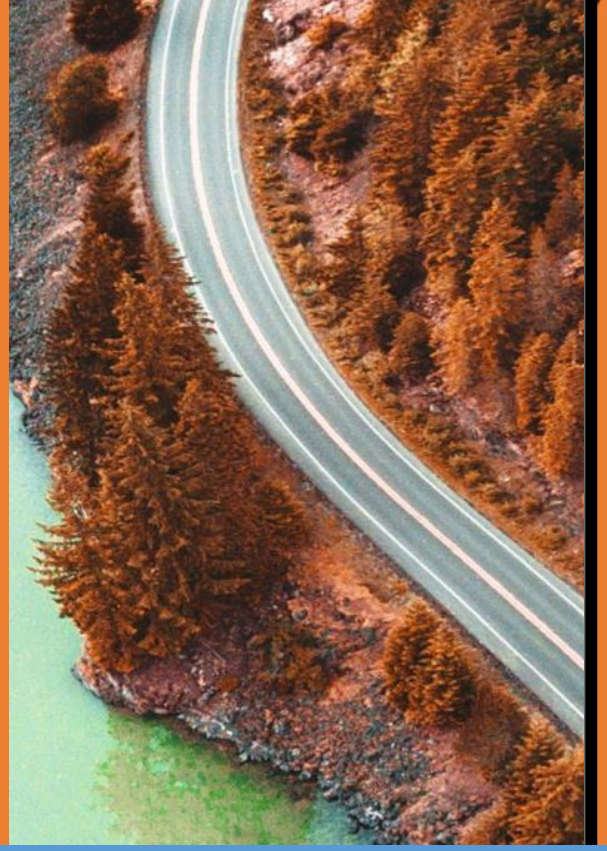
$\bar{y} = \bar{a} + b'\bar{x}$

- Univariate linear regression assumes that a linear equation is

  Using the above equation

$\bar{a} = \bar{y} - b'\bar{x}$

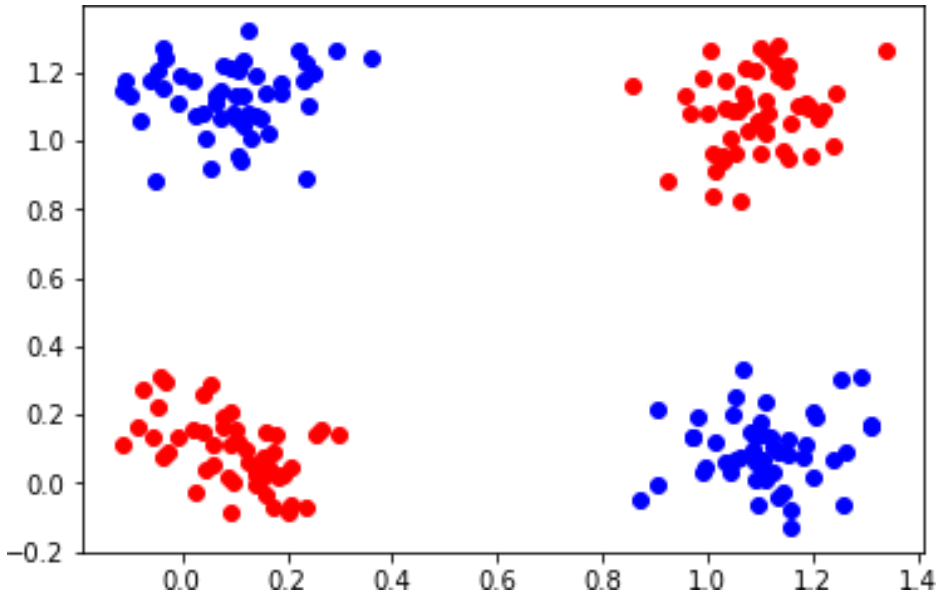




Non-linear models

XOR problem

Dataset



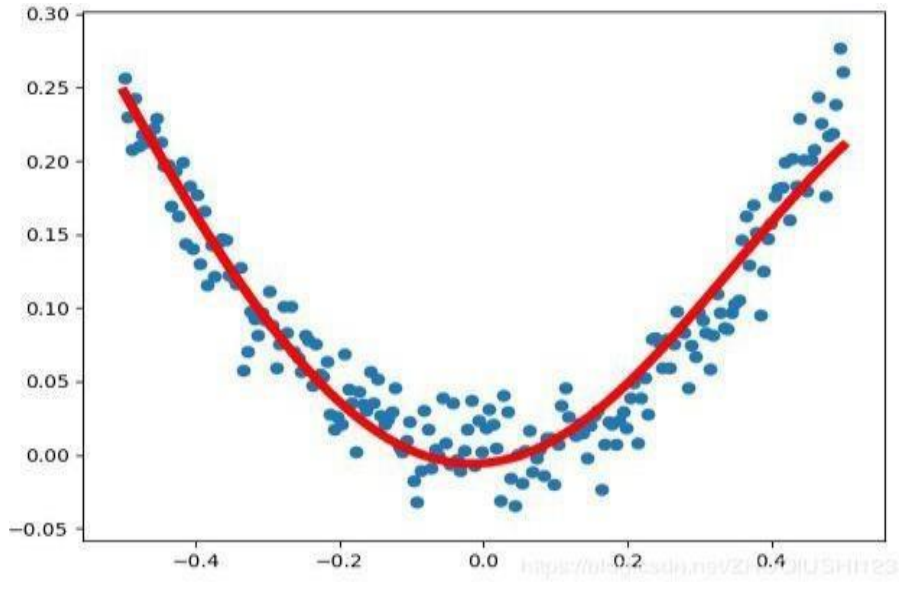


Figure: The XOR problem is a typical example of a dataset in which the classes arenot linearly separable.

Nonlinear Regression Types

1. **Transformable nonlinear models:** models involving a single predictor variable in which transforming Y, X or both results in a linear relationship between the transformed variables.

2. **Polynomial models:** models involving one or more predictor variables which include higher-order terms such as $B_{1,1}X_1^2$, $Y = B_0 + B_1X + B_2X^2 + B_3X^3$.

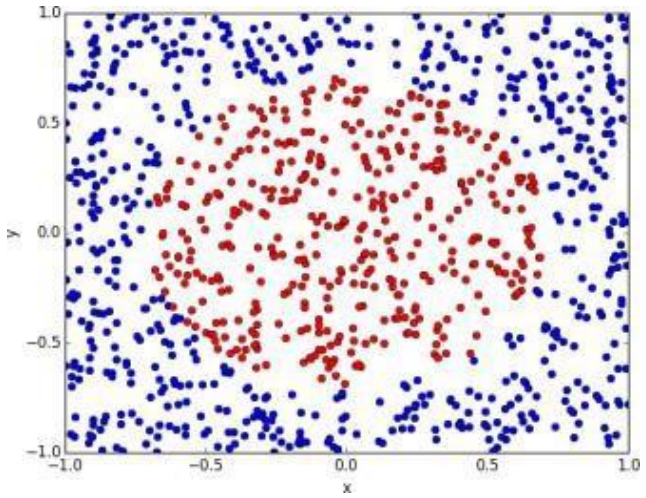
3. **Models that are nonlinear in the parameters**

Transformation of data

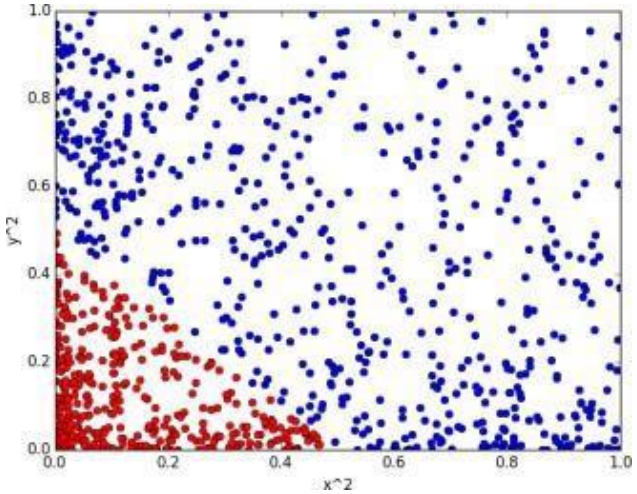
- Sometimes it can be easier to do data mining on a transformed variable than the original.
- Sometimes it can also be good to add a transformation of a variable as an extra variable.

Why transformations can be good

The problem is to find a classifier that given $\{x_1, x_2\}$ either classifies the point as red or blue.



(a) The original data



(b) The transformed data where all points $\{x_1, x_2\}$ are transformed to $\{x_1^2, x_2^2\}$

Figure: An example where transformation of the input variables makes the problem easier to solve

Thank you for listening